

LEARNING MADE EASY

Snowflake Special Edition

Building Applications with Snowpark

for
dummies[®]
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Python, Java, and
Scala in Snowflake

Scalable Data Pipelines
and AI/ML Models

Built-in Governance
and Security

Brought to you
by



Monica Mehta

About Snowflake

Snowflake enables every organization to mobilize their data with Snowflake's Data Cloud. Customers use the Data Cloud to unite siloed data, discover and securely share data, power data applications, and execute diverse AI/ML and analytic workloads. Wherever data or users live, Snowflake delivers a single data experience that spans multiple clouds and geographies. Thousands of customers across many industries, including 647 of the 2023 Forbes Global 2000 (G2K) as of October 31, 2023, use Snowflake Data Cloud to power their businesses. Learn more at snowflake.com.

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Table of Contents

INTRODUCTION 1

- About This Book 2
- Icons Used in This Book..... 2
- Beyond the Book..... 3

CHAPTER 1: Introduction to Modern Applications and Snowpark 5

- What Are Modern Applications? 6
- Exploring Use Cases for Apps 7
- Architecture of Applications..... 8
 - Data tier 8
 - Processing tier 9
 - Presentation tier 9
- Looking at the Benefits of a Cloud Data Platform for Applications..... 9
- Introducing Snowflake's Snowpark 10
 - Libraries 10
 - Runtimes 11
 - Processing of all data types 15

CHAPTER 2: Welcome to the Snowflake Data Cloud 17

- Compute Challenges for Applications 18
 - Multiple programming languages 18
 - Unstructured data 19
 - Security and governance..... 19
- Introducing the Snowflake Data Cloud..... 20
 - Data tier: Flexible architecture patterns 21
 - Processing tier: Language flexibility 22
 - UI tier: Streamlit in Snowflake 23
- The Benefits of the Data Cloud..... 23
 - Near-limitless scale and performance 24
 - Exceptional economic value 24
 - Inherent ease of use..... 24
 - Multicloud and cross-cloud flexibility..... 24
 - Baked-in security and governance 25
 - Unique collaboration options..... 26

CHAPTER 3: **What You Can Build with Snowpark**..... 27

 Building Data Pipelines..... 28

 Data transformations 29

 Custom business logic..... 29

 Data science and ML pipelines..... 29

 Building AI/ML Models..... 30

 Model development 31

 Model deployment and management 32

CHAPTER 4: **Getting Started with Snowpark**..... 33

 The Snowpark API 34

 Code Editors and IDEs 34

 Notebook Solutions for Data Science 34

 Snowflake Notebooks..... 34

 Hex notebooks 35

 Open-source notebooks 35

 Snowflake Python Worksheets 35

CHAPTER 5: **Six Benefits of Building Apps with Snowpark**..... 37

 One Platform for All Languages 38

 Bringing the Compute to the Data 38

 Built-In Governance and Security..... 39

 Increased Performance and Reduced Price..... 39

 Building with Streamlit in Snowflake 40

 Distribute and Monetize with Snowflake Native Apps and Snowflake Marketplace 41

Introduction

Modern applications are revolutionizing the world. These applications are not only transforming the way we interact with technology but also reshaping various sectors of society. They're harnessing the power of vast amounts of data and leveraging advanced artificial intelligence (AI) algorithms to provide personalized experiences, make accurate predictions, automate processes, and drive innovation. From healthcare to finance, and education to entertainment, modern applications are opening up new possibilities and changing our lives in ways we could never have imagined.

In healthcare, modern applications are being used to predict disease outbreaks, improve patient care, and accelerate drug discovery. In education, applications provide personalized learning experiences and early identification of students who need more help. Financial firms are using applications to assess risks, detect fraud, and optimize investor strategies. Entertainment companies are using applications to recommend content, analyze audiences, and create virtual reality experiences. These are just a few examples of how modern applications are changing the world — the opportunities are endless.

However, your organization may have experienced just how complex and time-consuming building data-intensive, AI-enriched applications can be. Unfortunately, data is often fragmented across many systems and in complex formats, requiring complex data pipelines and difficult integrations. Constructing accurate AI and machine learning (ML) models requires storing, processing, and analyzing large volumes of clean, accurate data. Developers often need to set up separate compute environments for the multiple programming languages needed to build robust applications. They also need to ensure that the data they collect and share is secured and governed so it doesn't get into the wrong hands. Rest assured, this book can help.

About This Book

The Snowflake Data Cloud, including Snowpark, helps developers overcome the challenges of building and deploying applications. The Snowflake Data Cloud is the best choice for application architecture because it simplifies development and operations by providing a unified, secure, and fully governed cloud environment for data storage, integration analysis, and other computing tasks.

Snowpark was designed to make building data pipelines and AI/ML models in Snowflake a breeze using programming languages such as Python, Java, and Scala without any data movement. If you're looking for faster, easier, and more cost-efficient application development, look no further than the Snowflake Data Cloud and Snowpark, which this book introduces to you in a straightforward and easygoing manner.

Whether you're a data scientist, data engineer, or developer, this book provides you with information about the following topics:

- » **Modern applications:** What are they? What are the use cases? What's the best architecture for building and deploying them?
- » **The Snowflake Data Cloud:** What is it? How does it support applications? What are the benefits of using it, and what can organizations achieve by building applications on it?
- » **Snowpark:** What is it? How and why should you build data pipelines with it? What are the best practices for and benefits of developing and deploying AI/ML models on it? How can you get started with Snowpark?

Icons Used in This Book

Throughout this book, the following icons highlight tips, important points to remember, and more:



TIP

Tips guide you to easier ways to perform a task or helpful advice as you build, deploy, and monetize apps.



REMEMBER

This icon highlights concepts worth remembering as you immerse yourself in all things Snowpark.



TECHNICAL
STUFF

The jargon beneath the jargon, explained.

Beyond the Book

Once you're ready to build applications quickly and cost-effectively with Snowpark, getting started is easy. You can begin development anywhere that can run a Python kernel. Simply install the Snowpark API and establish a connection with your Snowflake account. The Snowflake website offers a number of resources to help you get started with Snowpark quickly and easily. To learn more, visit snowflake.com/snowpark.

Disclaimer: Snowpark is the set of libraries and runtimes in Snowflake that securely deploy and process non-SQL code. Snowpark is an ever-evolving product. Features and capabilities described in this book may not be generally available, be different than described, or no longer exist at the time of reading.

- » Defining modern applications
- » Checking out modern app use cases
- » Describing the three-tier approach to application architecture
- » Revealing the benefits of using a cloud data platform for applications
- » Introducing Snowpark for the app processing tier

Chapter 1

Introduction to Modern Applications and Snowpark

The increasing market opportunities for data-intensive, artificial intelligence (AI)-enriched applications are driven by the explosion of data fueled by advances in wireless connectivity, compute capacity, and the proliferation of Internet of Things (IoT) devices. This is an exciting time for developers to deliver innovative, data-driven apps to customers in every industry, from healthcare to entertainment. But too many of them spend most of their efforts on unproductive tasks. They wrestle with reconciling different technologies, waste time managing complex infrastructures, and struggle to hire in-demand operations specialists.

THE ERA OF DATA

According to a 2023 report by Grand View Research, the global software market size was valued at \$583.47 billion in 2022 and is expected to grow at a compound annual growth rate (CAGR) of 11.5 percent from 2023 to 2030. The report is titled “Software Market Size, Share & Trends Analysis Report By Enterprise Size, By Vertical (BFSI, Retail), By Deployment (On-premises, Cloud), By Type (Productivity Software, Application Software), By Region, And Segment Forecasts, 2023 - 2030,” — in case you want to read more about it.

Additionally, market intelligence firm IDC predicted in their “Data Age 2025” report that the volume of data created each year will top 160 zettabytes (*that’s a trillion gigabytes*) by 2025 — a tenfold increase over the amount of data created in 2016. That report’s full title is “Data Age 2025: The Evolution of Data to Life-Critical, Don’t Focus on Big Data; Focus on the Data That’s Big.”

The good news is that Snowflake offers developers an easier, faster way to build, monetize, and deploy modern applications. But before we get into the solution, let’s start with the basics. This chapter defines applications (or apps in shorthand), describes the three-tier approach to application architecture, reveals the benefits of using Snowflake to build apps, and introduces Snowpark — Snowflake’s set of libraries and runtimes that securely deploy and process non-SQL code, such as Python, Java, and Scala.

What Are Modern Applications?

Chances are, you’ve already interacted with a modern application today. Modern apps are data-intensive and AI-enriched. They process large volumes of complex and fast-changing data from different sources, analyze it with AI/machine learning (ML) models, and present those insights so that customers or employees can make better decisions and perform smarter actions.

In other words, modern apps handle data collection, processing, and representation to provide value in the form of AI-enriched insights. Here are some examples:

- » **Marketing apps** can forecast customer behavior and product strategies.
- » **Health apps** can identify and predict disease, and personalize health recommendations.
- » **Transportation apps** can optimize shipping routes in real time.
- » **Security apps** help teams monitor, detect, and respond to events quickly.
- » **Climate apps** predict extreme weather events and enable municipalities to plan and respond quickly.

The possibilities are endless.

Modern apps have four characteristics:

- » **High data volume:** We're talking billions of images or millions of customer interactions.
- » **Complex data formats:** Data can be in a structured, semistructured, and unstructured format, such as text files, social media posts, or images and videos.
- » **Dynamic, ever-changing data:** The content and ingestion of data from external sources (such as web data via APIs, IoT devices, and other applications) can change and fluctuate wildly.
- » **AI-enriched:** Modern apps use AI/ML models for faster time to insights.

Exploring Use Cases for Apps

Here are five of the most common types of use cases for building applications that are data-intensive and AI-enriched:

- » **Customer 360:** With Customer 360, organizations can get a complete view of their customers by integrating data from various touchpoints and applying it in areas such as personalized marketing and customer service.

- » **IoT:** IoT apps ingest large volumes of time-series data from devices and sensors for near real-time analysis in areas such as security and personal health.
- » **Application health and security analytics:** Analytics are crucial to maintaining the performance, availability, and security of applications through use cases such as performance availability and compliance monitoring.
- » **ML and data science:** Apps can use ML algorithms for predictive analytics, image and speech recognition, fraud detection, and customer service to name just a few examples.
- » **Embedded analytics:** Embedded analytics is the integration of data analytics capabilities within business applications, allowing end users to access data insights.



TIP

Regardless of the type of application you're designing, building, or supporting, the best way to differentiate your offering is to select the right underlying architecture. Modern applications that deliver real-time value at massive scale require a modern data platform that's designed for highly performant customer experiences.

Architecture of Applications

The architecture of a modern application includes three physical tiers. The chief advantage of three-tier application architecture is that developers can develop, modify, and scale the tiers separately instead of changing the entire application.

Data tier

At the bottom is the data tier, sometimes called the database tier or persistence layer. This is where the data that feeds the processing tier is stored and managed. It includes the data storage and access mechanisms. Ideally, this tier can be scaled easily to allow for changes in the amount of data the application needs to handle at specific times.

Processing tier

In the middle is the *processing tier*, also known as the *business logic* or *compute layer*. This is where business logic for the application is defined — for example, through a specific set of business rules. This tier processes information that's collected in the presentation tier as well as information stored in the data tier through data transformations and ML models. The processing tier can also add, change, or delete data in the data tier. The most popular programming languages used in this tier are Python, Java, and Scala.

Presentation tier

On the top is the *presentation tier*, which houses the user interface. This is where the user interacts with the data. Its main purpose is to display information, communicate, and collect data from the end user. For example, the presentation tier of a retail application might include the mechanism for browsing merchandise, interacting with the shopping cart, and purchasing.



TIP

A major advantage of three-tier application architecture is an increased level of application security. Since the presentation and data tiers are separate, cyberattacks can be prevented by adding security features in the processing tier.

Looking at the Benefits of a Cloud Data Platform for Applications

A *cloud data platform* provides a unified, secure, and fully governed cloud environment for data store, integration analysis, and other computing tasks. It is the best choice for application architecture because it simplifies development and operations (DevOps) and lets you bring innovative applications to market faster and more efficiently.

The ideal modern cloud data platform for applications should include the five capabilities highlighted in Table 1-1.

TABLE 1-1 The Ideal Modern Cloud Data Platform

Capability	What It Does	Benefit
The capacity for near-unlimited resources	Separates compute from storage	Enables users to scale up and down automatically and work concurrently without impacting performance
Governed and secure collaboration	Eliminates the need to move or copy data internally or externally	Enables seamless and secure collaboration
A fully managed solution	Handles maintenance, administration, and other automated services	Decreases costs and valuable worker time otherwise spent on manual tasks
Built-in distribution capabilities	Publishes and offers apps to organizations that can easily find, try, and buy your apps	Increases app visibility and ease of monetization
Flexibility of deployment models	Runs the app in the provider's or consumer's account	Eliminates data exposure outside of secure environments

Introducing Snowflake's Snowpark

Now that you understand why it's critical to build applications on a cloud data platform, it's time to reveal why the best solution for processing data that feeds into applications through data transformations or AI/ML modeling is Snowpark on Snowflake.

Snowpark is the set of libraries and runtimes in Snowflake that securely enable developers to deploy and process non-SQL code, including Python, Java, and Scala. Thousands of organizations are accelerating development and performance of their data engineering and ML workloads with Snowpark for Python.

Libraries

On the client side, Snowpark consists of libraries that include the Snowpark API and Snowpark ML API. These are open-source libraries that work with any Python environment.

SNOWPARK CLIENT-SIDE LIBRARIES

Client-side libraries include:

- **Snowpark API:** The Snowpark API library includes the DataFrame API. The Snowpark DataFrame API brings deeply integrated, DataFrame-style programming and OSS-compatible APIs to the languages data practitioners like to use.
- **Snowpark ML API:** Snowpark ML includes the Python library and underlying infrastructure for end-to-end ML workflows in Snowflake. Snowpark ML unifies data pre-processing, feature engineering, model training, and integrated deployment into a single, easy-to-use Python library.

Runtimes

On the server side, runtimes include either Python, Java, and Scala in the warehouse model or Snowpark Container Services. Snowflake's virtual warehouses are compute clusters that host and run server-side contracts, such as user-defined functions (UDFs) and stored procedures for processing custom logic.

For Python developers, Snowpark's Python runtime makes it possible to write custom Python code through UDFs and stored procedures, which are deployed into Snowflake's secure Python sandbox. UDFs and stored procedures are two key components of Snowpark that allow developers to bring custom Python logic to Snowflake's compute engine, while taking advantage of open-source packages preinstalled from Anaconda in Snowpark.

For workloads that require the use of GPUs or other specialized hardware, custom runtimes and libraries, or hosting of long-running full-stack applications, Snowpark Container Services offers the ideal solution.

SNOWPARK SERVER-SIDE RUNTIMES

Server-side runtimes include:

- **Python, Java, and Scala in the warehouse model:** Runtime contracts include UDFs and Stored Procedures for Python that bring custom logic to Snowflake's compute engine while taking advantage of open-source Python packages pre-installed in Snowpark.
- **Snowpark Container Services:** This allows developers to register, deploy, and run container images in Snowflake-managed infrastructure.

UDFs

Custom logic written in Python runs directly in Snowflake using UDFs. These functions can stand alone or be called as part of a DataFrame operation to process the data. Snowpark takes care of serializing the custom code into Python byte code and pushes all of the logic to Snowflake, so it runs next to the data.

To host the code, Snowpark has a secure, sandboxed Python runtime built right into the Snowflake engine. Python UDFs scale out processing associated with the underlying Python code, which occurs in parallel across all threads and nodes, and comprising the virtual warehouse on which the function is executing.

There are several types of UDFs that can be used in Snowpark, including:

- » **Scalar UDFs:** Operate on each row in isolation and produce a single result
- » **Vectorized UDFs:** Receive batches of input rows as pandas DataFrames and return batches of results as pandas arrays or series
- » **User-Defined Table Functions:** Return multiple rows for each input row, return a single result for a group of rows, or maintain state across multiple rows

Stored Procedures

Snowpark Stored Procedures help developers operationalize their Python code and run, orchestrate, and schedule their pipelines. A stored procedure is created once and can be executed many times with a simple `CALL` statement in your orchestration or automation tools. Snowflake supports stored procedures in SQL, Python, Java, JavaScript, and Scala, so developers can easily create polyglot pipelines feeding applications.

Developers can use stored procedures in Snowpark to bundle the Python function and have Snowpark deploy it on the server side. Snowpark will serialize the Python code and dependencies into bytecode and store them in a Snowflake stage automatically. They can be created either as a temporary (session-level) or permanent object in Snowflake.

Stored procedures are single-node bound, which means transformations or analysis of data at scale inside a stored procedure should leverage the Snowpark DataFrame API or other deployed UDFs to scale compute across all nodes of a compute cluster.

Anaconda packages

One of the benefits of Python is its rich ecosystem of open-source packages and libraries. In recent years, open-source packages have been one of the biggest enablers for faster and easier development. To leverage open-source innovation, Snowpark has partnered with Anaconda for a product integration without any additional cost to the user beyond warehouse usage. Developers in Snowflake are now able to speed up their Python-based pipelines by taking advantage of the seamless dependency management and comprehensive set of curated open-source packages provided by Anaconda — all without moving or copying the data.

All Snowpark users can benefit from thousands of the most popular packages that are preinstalled from the Anaconda repository, including FuzzyWuzzy for string matching, H3 for geospatial analysis, and scikit-learn for ML and predictive data analysis. Additionally, Snowpark is integrated with the Conda package manager so users can avoid dealing with broken Python environments because of missing dependencies.

Snowpark also fully supports dbt, one of the most popular solutions for data transformation today. It supports a SQL-first

transformation workflow, and in 2022, dbt introduced support for Python. With dbt's support for both SQL and Python, users can write transformations in the language they find most familiar and fit for purpose. And dbt on Snowpark allows analyses using tools available in the open-source Python ecosystem, including state-of-the-art packages for data engineering and data science, all within the dbt framework familiar to many SQL users.

Snowpark-optimized warehouses

Snowpark-optimized warehouses have compute nodes with 16x the memory and 10x the local cache compared with standard warehouses. The larger memory helps unlock memory-intensive use cases on large data sets such as ML training, ML inference, data exports from object storage, and other memory-intensive analytics that could not previously be accommodated in standard warehouses.

Snowpark-optimized warehouses also inherit all the benefits of Snowflake virtual warehouses:

- » **Fully managed:** Snowflake oversees the maintenance, security patching, tuning, and delivery of the latest performance enhancements transparently.
- » **Elastic:** Elastic scaling of compute supports virtually any number of users, jobs, or data with multitenant security and resource isolation.
- » **Reliable:** Industry-leading SLA is consistently upheld.
- » **Secure:** Governance controls are applied across all workloads without trade-offs.

Snowpark Container Services

Snowpark Container Services is a new Snowpark runtime option that enables developers to effortlessly deploy, manage, and scale containerized workloads (jobs, services, service functions) using secure, Snowflake-managed infrastructure with configurable hardware options, such as GPUs. This new runtime eliminates the need for users to deal with complex operations of managing and maintaining compute and clusters for containers.

Moving governed data outside of Snowflake would expose it to security risks. With containers running in Snowflake, there is no

need to move governed data outside of Snowflake to use it as part of the most sophisticated AI/ML models and apps (whether developed for internal or for commercial uses). Snowpark Container Services is available for workloads that require use of GPUs, custom runtimes/libraries, or the hosting of long-running full-stack applications.

The containers, built and packaged by developers using their tools of choice, can include code in any programming language (for example, C/C++, Node.js, Python, R, React, etc.). In addition, Snowpark Container Services provides a simple, unified experience for the end-to-end lifecycle of containerized applications and AI/ML models. Other solutions require you to manually stitch together a container registry, container management service, and compute service, plus require you to manage your own separate tools for observability, data connectivity, security, and so on.

Snowflake Native Apps provide the building blocks for app development, distribution, operation, and monetization all within Snowflake's platform. Snowpark Container Services can be used as part of a Snowflake Native App to allow developers to build and distribute sophisticated full-stack apps that run entirely in their end customer's Snowflake account.

Processing of all data types

Developers can ingest all types of data using Snowflake's platform, including streaming or batch ingestion of structured, semistructured, or unstructured data. The potential of unstructured data for analytics and applications has remained largely untapped.

Snowpark allows users to easily process and derive insights from unstructured data from files such as images, videos, and audio. Python developers can easily take advantage of the Python ecosystem of open-source packages, such as PyPDF2 for PDF processing, or OpenPyXL to read Excel files.

IN THIS CHAPTER

- » Detailing the challenges of application development
- » Identifying the unique attributes of the Data Cloud
- » Exploring how the Data Cloud supports applications
- » Revealing what organizations can achieve from building apps with Snowflake

Chapter 2

Welcome to the Snowflake Data Cloud

Application development is a complex and time-consuming process. The good news is that new tools are making the task of building and deploying applications much easier. According to industry reports, generative artificial intelligence (AI) can help developers complete coding tasks in 30 to 50 percent of the time. But legacy systems can't handle the large amounts of data that are required to build AI applications, making it difficult to gather, analyze, and share data across entities.

To use advanced tools that make it easier to build applications, you need a scalable, flexible data platform and application architecture. Here's where Snowflake comes in. The Snowflake Data Cloud provides the platform to help developers build and deploy applications quickly, easily, and cost-effectively.

With this technology foundation, organizations can overcome compute challenges for applications and even build innovative new types of applications.

Compute Challenges for Applications

One prediction for how data will be used in 2025 is that it will be “embedded in every interaction and process.” That’s according to Snowflake’s Data Trends 2023 Report. As the amount of data grows exponentially, it becomes increasingly difficult to manage and process in a timely manner. This requires the development of new AI and machine learning (ML) algorithms that can scale to search and process massive amounts of data.

However, modern applications often have complex workflows that can be difficult to manage, scale, and secure, and they require advanced infrastructure to support them. Modern applications also tend to have intensive resource demands, requiring large amounts of compute power, networking bandwidth, and graphics capabilities.

In this section, we describe three specific compute challenges for developers building and deploying applications — handling multiple programming languages, processing unstructured data, and managing complex data governance — and then in the next section show how these challenges can be overcome.

Multiple programming languages

Development teams often need to use multiple programming languages to build robust applications. SQL and Python are among the most popular languages in the modern data stack processing tier, where data is processed and transformed, and where ML models are trained and deployed. While SQL is the long-standing database language for querying and transforming data, Python has emerged as the preferred programming language for developers as more advanced logic such as AI and ML becomes part of the application.

The challenge arises when developers need to set up and manage separate compute environments for each language. Moving data across compute environments often requires converting, egressing, scanning, and loading data from its source. And developers often need to stitch multiple tools together to finish a single analysis when using multiple languages. Even for those confident in both languages, running separate clusters for each environment can be frustrating and time-consuming.

Unstructured data

Unstructured data can provide a rich source of information for application developers, leading to deeper insights about user behavior and user experience. Unstructured data doesn't have a predefined structure and can include call logs, chat transcripts, contracts, and sensor data. It could be in any file format, such as doc, pdf, jpeg, mp3, or mp4. It could also be in industry-specific formats, such as raster data for geospatial, .las files for well log data, or DICOM for medical.

Because the limitations of legacy technology made processing solutions very complex and expensive, the potential of unstructured data for analytics and applications has remained largely untapped. Historically, organizations have relied on structured tables and semistructured data such as JSON and XML for analytical workloads and applications. For developers, unstructured data is often hard to manage and not easy to collect, search, or query. The sheer size and complexity of information without a predefined schema also make it challenging to extract actionable insights.

Security and governance

Developers must focus on security and governance to ensure their application aligns with business goals and complies with external regulations. Strong application security is critical because it protects software application code and data against cyberthreats. Governance provides a formal framework for achieving measurable progress toward strategic objectives, maintaining compliance standards, and protecting data privacy and security.

But keeping up with the evolving threat landscape and the adoption of nimbler, more incremental software development frameworks can be difficult and expensive. In addition, data is often siloed and fragmented throughout an organization, but strong security and governance requires a centralized repository for security-relevant data.

Finally, manual processes of reviewing alerts and validating controls slows down reaction time to incidents. Organizations need a data platform that can ensure multilevel protections, extending the security supplied by cloud storage vendors.

DATA DIFFICULTIES FOR DEVELOPERS

Data presents a number of challenges for application developers, who must:

- Break through organizational silos that prevent access to a single view of data.
- Unify various formats of structured and unstructured data from multiple sources.
- Set up and manage separate compute environments for each language.
- Store and process massive volumes of data quickly and easily to construct accurate AI models.
- Ensure that the data they collect and share complies with regulatory guidelines.
- Safeguard the security of the data and make sure it doesn't get into the wrong hands.

(And they must do it all while managing costs!)



TIP

Regardless of the programming language they use or the type of data they collect, application developers need to ensure the application architecture is reliable, can scale to meet demand, and has built-in security and governance.

Introducing the Snowflake Data Cloud

To solve these challenges, more and more organizations are turning to the Snowflake Data Cloud. That's because the Data Cloud is a global network where thousands of organizations mobilize data with near-unlimited scale, concurrency, and performance. Inside the Data Cloud, organizations can unite their siloed data, easily discover and securely share governed data, and execute diverse analytic workloads.

Wherever data or users live, the Data Cloud delivers a unified and seamless experience — even when data and workloads span multiple public clouds. The Data Cloud enables discovering, managing, and sharing data among business units, suppliers, other business partners, and customers. It also offers live access to ready-to-use data and applications that power innovative business solutions from hundreds of providers in Snowflake Marketplace.

Snowflake's flexible architecture lets you develop and scale applications without operational burden. Snowflake's platform supports three robust tiers or layers (introduced in Chapter 1) that work together to make the Data Cloud an ideal platform for modern applications.



REMEMBER

For your application storage or data tier, Snowflake can handle all types of data with automatic scalability and reliability. For the processing tier, Snowflake's elastic compute infrastructure allows developers to write code in their preferred language for pipelines and AI/ML models and run it directly on Snowflake. For the presentation tier or user interface layer, Snowflake offers multiple options to build and host front-end applications using the language of your choice.



TECHNICAL
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Snowflake's platform is a unified architecture that allows you to integrate any type of data — structured, semistructured, or unstructured — from a wide range of data sources and use it to power your applications, from search components to AI features. Snowflake's platform, delivered as a fully managed service, features storage, compute, and global services layers that are physically separated but logically integrated. Data workloads scale independently from one another, making it an ideal platform for scaling applications without resource contention or performance degradation.

Data tier: Flexible architecture patterns

Snowflake's single platform enables customers to work with more than SQL — it powers a full spectrum of use cases against many architecture patterns. That means architectures can be built based on what works best for an organization, even if that changes over time. With this flexibility, customers can power any use case while staying true to the core tenets of the Snowflake platform: strong security and governance, excellent performance, and simplicity.

Snowflake supports a wide range of data formats, architecture patterns, and use cases — all on a single, secure, and governed platform — including any of the following:

- » A data warehouse for highly optimized SQL and business intelligence use cases
- » A data lake for structured, semistructured, or unstructured data
- » A data mesh for a team to manage its own data and adhere to standards (rather than a central data team)
- » A data lakehouse for SQL analytics and unstructured data processing together

With Snowflake, you have two data storage options:

- » Use Snowflake as your central data repository and supercharge performance, querying, security, and governance with the Snowflake Data Cloud.
- » Store your data in AWS S3, Azure Data Lake, Google Cloud Storage, or S3-compatible storage and use Snowflake to add a governance layer on top and speed up data transformation and analytics using Snowflake compute.

Processing tier: Language flexibility

Snowflake provides full support for programming languages such as Python, Java, and Scala. For development and processing in other languages, Snowpark enables Snowflake to meet the programmability needs for all developers, regardless of which languages you use to code your applications. Snowpark's set of libraries and runtimes in Snowflake securely deploy and process programming languages such as Python, Java, and Scala.

To transform data or build features for AI/ML models, you can use DataFrame-style APIs that help you interact with Snowflake's SQL engine with familiar programming. To process custom code inside warehouses, which consist of Snowflake's elastic compute clusters, developers can package their code with User-Defined Functions (UDFs) and Stored Procedures.

In the case of Python code, third-party open-source libraries can be securely and efficiently executed using the integrated Anaconda package repository and manager. All compute is automatically

pushed to the Snowflake engine for the heavy lifting to enable use cases such as data transformations, ELT/ETL pipelines, ML development, and ML deployment. With Snowflake, it all happens in a single, easy-to-use platform with superior price performance and near-zero maintenance.

Snowpark supports two types of compute: warehouses and containers. Most data processing logic can be efficiently executed in Snowflake using the warehouse compute that supports Python, Java, and Scala with standard compute or high-memory compute (Snowpark-optimized warehouse).

Developers that require programming language or hardware flexibility can build and deploy using Snowpark Container Services, Snowflake's fully managed container offering. Containers can be built and packaged by developers using their tools of choice, can include code in any programming language (for example, C/C++, Node.js, Python, R, React, etc.), and can be executed using a wider range of configurable hardware options, including GPUs.

UI tier: Streamlit in Snowflake

Building user interfaces does not require advanced front-end development expertise. Snowflake makes it easy for developers to build and deploy interactive interfaces for their applications using the native integration of the Streamlit Python library. For teams that need ultimate flexibility and employ experienced front-end developers, Snowpark Container Services supports the hosting and execution of any programming language.

The Benefits of the Data Cloud

The Snowflake Data Cloud is the natural platform for applications. Snowflake's unique architecture solves many of the challenges that developers face when they build and deploy applications, and gives them the power to create modern applications that are data-intensive and AI-enriched.

Snowflake customers are creating full-stack modern apps, including apps that create advanced visualizations for near real-time risk awareness, feature visual self-service analytics, and have natural language chat interfaces that run in the end user's

data platform. In this section, we highlight some of the benefits of building applications on Snowflake.

Near-limitless scale and performance

Snowflake is architected for cloud-scale computing in terms of data volume, computational performance, and concurrent workload execution. This unique cloud-built architecture enables organizations in the Data Cloud to spin up new workloads and dynamically provision those workloads with as little or as much compute and storage capacity as they need. There are no hidden capacity limits, and there is no fear of resource contention with other concurrent data workloads.

Exceptional economic value

Snowflake's cost-effective utility model ensures organizations in the Data Cloud pay only for the capacity they use, in one-second increments. Capacity is provisioned automatically, based on user-defined thresholds, allowing customers to focus on consumption while the platform manages the resources they need. Because Snowflake contracts with the public cloud vendors for immense capacity at a considerable scale, customers receive these cloud services at exceptional economic value via Snowflake.

Inherent ease of use

Snowflake's platform is dramatically more straightforward to use than previous generations of cloud data platforms and database-as-a-service offerings. Because the platform doesn't require expert knowledge to run, your database administrators can move beyond manual tuning and configuration chores, and your developers can focus their expertise on creating and refining data models, extracting new insights, and deriving business value from a wide range of data.

Multicloud and cross-cloud flexibility

The Data Cloud spans the world's most popular cloud service providers (CSPs): Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP). You no longer have to standardize on one of them. You can leverage multiple regions and clouds according to your business needs and seamlessly share data from one region or cloud to another. For example, a large multinational

organization with a multicloud strategy can use the Data Cloud to operate seamlessly across multiple cloud provider regions to serve its worldwide customer base and enhance its disaster recovery strategy. Similarly, a data provider can leverage Snowflake's global infrastructure to share data with other organizations in many geographies and across multiple clouds.



TIP

Because Snowflake completely abstracts the underlying public cloud platform, users and administrators don't need any cloud-specific expertise. They never have to work with third-party cloud services directly, nor do they receive a separate bill for public cloud usage.



REMEMBER

Snowflake is provided as a fully managed service that runs completely on the CSP's infrastructure. Snowflake designed the platform to be cloud agnostic, and a Snowflake account can be hosted on any of the following cloud platforms: AWS, Microsoft Azure, and GCP. One unified code base spans all three public cloud services, which means customers can seamlessly move data and workloads among them yet interact with one cohesive platform interface. Being cloud agnostic is a decisive advantage when serving a worldwide user base or formulating a global disaster recovery strategy.

Baked-in security and governance

A strong data governance and compliance framework is critical to extracting the analytical value of data. The Data Cloud allows you to store all your data while providing fast, governed, and secure access to that data. Industry-leading cybersecurity practices extend across all data and all workloads. Multilevel security includes data encryption and associated key management services, role-based access controls, object-level permissions, and robust database security. Furthermore, the Data Cloud adheres to regulatory and data privacy policies to ensure the correct handling of sensitive data.



REMEMBER

Although new privacy requirements are still emerging in many areas, Snowflake is continually evolving to help customers comply with national and industry-specific regulations, and to avoid costly violations of those regulations.

Unique collaboration options

Snowflake's unique architecture provides the near-unlimited scale, concurrency, and performance for organizations to collaborate with data across their enterprises and ecosystems. Whether sharing data or full-stack applications, the Data Cloud raises the bar of what is possible.

You can also establish entirely new revenue streams and collaborate with business partners in new and productive ways with capabilities such as the Snowflake Native App Framework, which allows developers to build, distribute, and monetize full-stack apps that run within the end customer's Snowflake account.



TIP

The Snowflake Native App Framework provides the building blocks for app development, distribution, operation, and monetization all within Snowflake's platform. It gives providers the capability to build a new type of application — a Snowflake Native App — that's distributed and monetized on Snowflake Marketplace and runs inside the end customer's Snowflake account.



REMEMBER

Developers can use the flexibility and programmability of Snowpark's libraries and runtimes when building Snowflake Native Apps. This includes Snowpark Container Services, an additional Snowpark runtime that enables developers to manage and deploy container images within Snowflake-managed infrastructure using configurable hardware options including NVIDIA AI GPUs.

- » Using Snowpark for data engineering use cases
- » Revisiting Snowpark components
- » Revealing what makes building AI/ML models successful
- » Detailing model development and model deployment with Snowpark

Chapter 3

What You Can Build with Snowpark

Snowpark, the set of libraries and runtimes in Snowflake that securely deploy and process Python and other programming languages, enables all data users to bring their work to the Snowflake Data Cloud. Snowpark allows data engineers, data scientists, and data developers to code with their language of choice to execute pipeline, machine learning (ML) workflow, and app logic faster and more securely, in a single platform. With Snowpark, users coding any language, including Python, Java, and Scala, will be able to benefit from the performance, elasticity, and governance of the Snowflake platform.

This chapter examines how you can build better data pipelines and ML models with Snowpark.

Building Data Pipelines

Data pipelines feeding applications can be complex for two main reasons:

- » Multiple tools and infrastructure for each programming language

One of the primary reasons to support multiple languages is that data engineering is a collaborative effort across teams. Data analysts may prefer SQL and GUI-based tools like business dashboards. In contrast, data scientists generally prefer to prepare data using notebooks and Python, and data engineers and developers need additional tools to tackle complex code and programming constructs. Data often needs to travel across these different systems to make pipelines work, leading to complex architectures that can jeopardize security and impair data governance.

- » Significant manual effort and maintenance overhead

Working with complex data processing infrastructures typically requires a lot of management and troubleshooting. As a result, data engineers are often spread thin, spending most of their time maintaining and fixing pipelines.

Snowflake natively supports SQL processing, and with Snowpark's libraries and runtimes, data engineers can also effortlessly build simple, governed, and fast pipelines with other popular programming languages, including Python, Java, and Scala. Snowpark offers data engineers three crucial benefits:

- » A single platform that supports multiple languages without any external compute
- » Superior governance
- » Faster, cheaper, and more resilient pipelines

Snowflake's intelligent infrastructure eliminates the complexity that data engineers often face — it just works, so they can focus on what matters.

Data engineers commonly use Snowpark for three use cases: data transformation/feature engineering, custom business logic, and data science and ML pipelines.



TIP

Customers see a median of 3.5x faster performance and 34 percent cost savings with Snowpark over managed Spark.

Data transformations

Snowpark can be used to transform data stored in Snowflake or S3-compatible data sources using popular programming languages such as Python, Java, and Scala. This can include data engineering tasks such as data cleaning, data normalization, and data aggregation. All data transformations can then be packaged as Snowpark stored procedures to operate and schedule jobs with Snowflake Tasks or other orchestration tools.

Data engineering teams can use Snowpark to transform raw data into modeled formats regardless of type, including unstructured data such as images, videos, and audio.

Custom business logic

Users can leverage Snowpark's User Defined Functions (UDFs) to streamline architecture with complex data processing and custom business logic written in Python or Java in the same platform running SQL queries and transformations. There are no separate clusters to manage, scale, or operate.

Data science and ML pipelines

Data teams can use the integrated Anaconda repository and package manager to collaborate in bringing ML data pipelines to production. Trained ML models can also be packaged as a UDF to run the model inference close to data, enabling faster paths from model development to production.



REMEMBER

Snowpark enables pipelines with better price performance, transparent costs, and less operational overhead thanks to Snowflake's unique multicloud shared data architecture. Snowflake is a single integrated platform that delivers the performance, scale, elasticity, and concurrency that today's organizations require.

SNOWPARK COMPONENTS

As a refresher, developers interact with Snowpark using libraries and runtimes:

- **Snowpark libraries:** On the client side, Snowpark consists of libraries including the Snowpark API for DataFrame transformations and the Snowpark ML API for end-to-end artificial intelligence (AI)/ML development and operations.
- **Snowpark runtimes:** On the server side, runtimes include Python, Java, and Scala in warehouse compute and support for any language in Snowpark Container Services. In warehouses, developers can leverage UDFs and stored procedures (sprocs) to bring in and run custom logic. Snowpark Container Services is available for workloads that require use of GPUs, custom runtimes/libraries, or the hosting of long-running full-stack applications.

Building AI/ML Models

Many applications are built on AI/ML models, but taking these models to production requires multiple teams. Doing so in a scalable and repeatable way isn't easy when each team manages its own copies of data spread out over its own processing infrastructure. Snowflake solves these challenges by providing these three key benefits:

- » **Centralized, governed data:** Teams can access a single true source of data in your organization, with easy access to structured and unstructured data from other external sources — whether it's from another business unit, a partner, or a third-party source — without manual transfers using Snowflake Marketplace.
- » **Flexible, elastic compute engine:** Multiple teams can access Snowflake's compute engine concurrently to process data for features, models, and applications. They can also bring advanced analytics such as AI and ML to the data without having to move it to external, ungoverned environments.
- » **Choice of tools:** Teams can interact with Snowflake directly from the Snowflake UI or from another tool. Snowflake

provides developers with the flexibility to use their tool of choice while ensuring the processing and data stays within Snowflake's secure and governed boundaries.

To access, visualize, and process data as part of their ML workflows, data scientists can leverage their favorite programming languages, such as Python, using Snowpark in Snowflake.

For Python users, Snowpark ML includes a client-side library and underlying infrastructure for end-to-end ML workflows in Snowflake. Snowpark ML unifies data preprocessing, feature engineering, model training, and integrated deployment into a single, easy-to-use Python library. Snowpark ML includes the Modeling API for data preprocessing and model training, as well as the Snowpark ML Operations capabilities, including the Snowpark Model Registry, for model deployment and management.



Snowflake's infrastructure includes Snowpark-optimized warehouses — which have 16x the memory and 10x the local cache compared with standard warehouses — and are great for memory-intensive operations including ML model training and inference.

Model development

Model development refers to the process of designing, training, and evaluating an ML model. This involves selecting the appropriate algorithms, preparing the data, and tuning the model's parameters to achieve the best performance.

Snowpark ML Modeling API includes a collection of Python APIs for preprocessing data and training models. By using Snowpark ML Modeling to perform these tasks within Snowflake, you can:

- » Transform your data and train your models without moving your data out of Snowflake.
- » Work with popular ML frameworks similar to those you're already familiar with, such as scikit-learn and XGBoost, directly in Snowflake.
- » Keep your ML pipeline running within Snowflake's security and governance frameworks.
- » Benefit from the performance and scalability of Snowflake's compute infrastructure.

Model deployment and management

Model deployment refers to the process of integrating a trained ML model into a production environment so that it can be used to make predictions on new data.

Snowpark ML has capabilities for ML Operations (MLOps), including the Snowpark Model Registry, which enables scalable, secure deployment and management of models in Snowflake. This registry provides centralized publishing and discovery of models to streamline collaboration as part of the process where data scientists hand off successful experiments to ML engineers. Models managed in the Snowpark Model Registry can be deployed to production securely on elastic Snowflake infrastructure, including Snowflake warehouses and Snowpark Container Services.

Central management and governance of ML artifacts and metadata is critical to scalable and robust MLOps. By having a Snowflake native solution that can manage any ML model, whether trained inside or outside of Snowflake, Snowflake customers can streamline the process of running multiple models in production — all while having scalable access controls that democratize access to the models for SQL users and not just ML experts.



REMEMBER

Snowflake helps organizations maximize the value of data, including unstructured data and third-party data. Regardless of the language used, Snowflake's intelligent infrastructure handles scaling and performance tuning, allowing data scientists to spend more of their time building models.

With Snowflake, you can:

- » Publish, discover, and share models with a central and governed view of model artifacts and metadata.
- » Effortlessly deploy registered models for inference using scalable Snowflake infrastructure.
- » Democratize access with the ability to expose models and results to coders and SQL users

- » Minimizing the learning curve with the Snowpark API
- » Building code using code editors and IDEs
- » Building ML models using notebooks
- » Test-driving Snowpark with Snowflake Python worksheets
- » Exploring Snowpark online resources

Chapter 4

Getting Started with Snowpark

Are you ready to build applications quickly and cost-effectively with Snowpark? It's easy to get started because Snowflake minimizes the learning curve by eliminating the need for a separate tool. You can begin development anywhere that can run a Python kernel. Simply install the Snowpark API and establish a connection with your Snowflake account.

Snowpark gives developers the flexibility to tailor their workflow to their preferences. It supports many development interfaces, including code editors and integrated development environments (IDEs), Hex and open-source notebook solutions, and Snowflake Notebooks and Snowflake Python worksheets in the Snowflake web interface.

The Snowpark API

The Snowpark API is a library, including a DataFrame API, that allows developers to query and process data at scale in Snowflake. The Snowpark DataFrame API enables developers to write queries and data transformations using familiar DataFrames while benefiting from Snowflake's processing performance, security, and scalability.

Code Editors and IDEs

Many developers prefer to build using code editors and IDEs. These offer capabilities such as local debugging, autocomplete, and integration with source control. Snowpark works well in VS Code, IntelliJ, PyCharm, and other tools. Code editors and IDEs are a great choice for a rich development and testing experience for building pipelines.



TIP

VS Code also works with a Jupyter extension that provides a notebook experience within the editor. This brings breakpoints and debugging into the notebook experience without requiring separate management of the Jupyter container or runtime.

Notebook Solutions for Data Science

Notebooks are a popular choice for building machine learning (ML) models in Snowpark. They enable rapid experimentation using cells, which means you can write code in one cell, run it, and see the results immediately without having to run the entire application. This makes it easier to test small parts of your code and debug issues. You can run a variety of notebook solutions with Snowpark.

Snowflake Notebooks

Snowflake Notebooks is a new development interface that offers an interactive, cell-based programming environment for Python and SQL users to explore, process, and experiment with data in Snowpark. Snowflake's built-in notebooks allow developers to write and execute code, train and deploy models using Snowpark

ML, visualize results with Streamlit chart elements, and much more — all within Snowflake's unified, secure platform.

And, because the Notebook is natively integrated into Snowflake's role-based access controls (RBAC), it's easy to securely share and collaborate on your code and results without compromising the security of any enterprise data. For data science and ML, the cell-based layout in Snowflake Notebooks unlocks experimentation and exploration tasks, as developers can write and execute code, visualize results, capture notes, and share insights all in one place.

Hex notebooks

Snowpark is integrated with Hex's notebooks, allowing for a powerful and flexible coding experience. Hex was built to work with Snowflake's Data Cloud. Its unique architecture and tight integrations let users move beyond the limits of traditional notebooks, and modernize their workflows for the cloud. Most if not all data processing can be pushed down to the Data Cloud, taking advantage of Snowflake's scalable compute and advanced caching. Discovering insights and building apps on data stored in Snowflake using Hex's analytics workspace takes just minutes.

Hex provides first-class support for Snowpark. Integrating with Hex's notebook UI provides an elegant front end for data scientists and analysts to connect to data in the Data Cloud, explore and analyze SQL and Python, build interactive data apps, and share them broadly.

Open-source notebooks

Open-source notebooks such as Jupyter can be run locally while connected securely to Snowflake to execute data operations. This lets developers interact with their data directly and see their queries in real time. Any machine running containers or Python can build data pipelines or ML models feeding apps using Snowpark.

Snowflake Python Worksheets

Snowflake Python worksheets are a quick and simple way to try Snowpark directly in Snowsight, the Snowflake web interface, for data manipulations and transformations. These worksheets

provide autocomplete for the Snowpark session and can run directly from the browser as a stored procedure. You can use third-party packages listed in the Snowflake Anaconda channel or import your own Python files from stages to use in scripts. If you're interested in trying Snowpark for free, the easiest way to get started is through the 30-day free trial using the Snowflake Python worksheets here:

<https://signup.snowflake.com/?lab=getStartedWithSnowparkInPythonWorksheets>

SNOWPARK ONLINE RESOURCES

The Snowflake website offers a number of resources to help you get started with Snowpark quickly and easily:

Snowpark Quickstarts offer several tutorials, such as “Getting Started with Snowpark for Python and Streamlit” or “Intro to Machine Learning with Snowpark for Python.” Visit <http://quickstarts.snowflake.com>.

Snowpark Developer Guides are available for each of the three supported programming languages: Java, Python, and Scala. Visit <https://docs.snowflake.com/en/developer-guide/snowpark/index> for more information.

The Snowpark homepage offers a wealth of information on all things Snowpark: snowflake.com/snowpark.

To get started with Snowpark, you need a Snowflake account and read/write access to a database. If you do not have a Snowflake account, you can sign up for a free trial, which doesn't even require a credit card!

<https://signup.snowflake.com/?lab=getStartedWithSnowparkInPythonWorksheets>

IN THIS CHAPTER

- » Streamlining architecture complexity with one platform
- » Processing code next to the data
- » Running apps in a governed, secure environment
- » Enjoying improved performance while keeping costs down
- » Optimizing app distribution
- » Maximizing monetization

Chapter 5

Six Benefits of Building Apps with Snowpark

Imagine you were building an application on a cloud platform that lets you run multiple programming languages without needing different processing engines. You could process your code next to the data in one governed, secure environment rather than exporting it to other locations. And you could build, deploy, distribute, and monetize your application natively on one scalable platform. What price and performance gains would you be able to achieve? How much more quickly would you be able to bring your app to market and serve your customers?

Snowpark, the set of libraries and runtimes that securely deploy and process programming code such as Python, Java, and Scala in Snowflake, opens data programmability to all developers, regardless of which languages you use to code your applications. You can code in your familiar constructs and run directly inside Snowflake's elastic compute engine, with no other processing system needed. And you can do everything from build to monetize apps on Snowflake's secure, governed platform. This chapter

details the many ways organizations can benefit from building the underlying application processing tier — pipelines and machine learning (ML) models — with Snowpark.

One Platform for All Languages

Developers often need to use multiple programming languages to build robust applications. Architecture complexity increases significantly when different teams use different languages across multiple processing engines. Snowpark streamlines architectures by natively supporting programming languages of choice — including Python, Java, and Scala — all in Snowflake without the need for separate processing engines. Instead, Snowpark brings all teams together to collaborate on the same data in Snowflake’s single platform. This can lead to more efficient collaboration and development productivity.



REMEMBER

Snowpark includes runtimes for Python, Java, and Scala in addition to Snowpark Container Services, which can accommodate code in any programming language of choice (for more on this, see Chapter 1).



TIP

Python adoption continues to surge with the rapid adoption of advanced analytics and ML in the enterprise. According to Snowflake’s Data Trends 2023 report, in February 2023, Python accounted for nearly 88 percent of all jobs run on Snowpark.

Bringing the Compute to the Data

One of the top four trends identified in Snowflake’s Data Trends 2023 report was that state-of-the-art companies are bringing their work and artificial intelligence (AI) to the data, not vice versa.

Snowpark lets developers write and execute code where the data is securely stored and governed. With Snowpark as the processing layer, developers can build and run applications directly on data already in Snowflake. In addition, Snowflake can automatically scale compute resources up and down for virtually unlimited concurrency without impacting performance or having to reshuffle data.

As a result, developers can build and scale applications powered by Snowflake without worrying about capacity planning or costly overprovisioning. Snowflake also provides a set of tools to write code, check that it works correctly, and put it into production — which can help accelerate the release of new applications or features.



REMEMBER

As we've mentioned elsewhere in the book, developers can use Snowpark to write code in their preferred language to process and analyze structured, semistructured, and unstructured data governed in Snowflake.

Built-In Governance and Security

Another trend highlighted in the Data Trends 2023 report was that organizations are paying increased attention to governance. One way to see how much attention is being paid to data protection is to look at how often data masking policies are set to limit access to the right users at the right times. Across the Snowflake Data Cloud in FY 2022, the number of applied masking policies grew more than 205 percent according to Snowflake's Data Trends 2023 report.

Enterprise-grade governance controls and security are built into Snowpark. For example, Snowpark's secure design isolates data to protect the network and host from malicious workloads, and gives administrators control over the libraries that developers execute. Developers can build confidently, knowing data security and compliance measures are incorporated and consistent.

Increased Performance and Reduced Price

Snowpark customers see significant price and performance benefits over other managed and open-source solutions, thanks to Snowflake's engine and the elimination of data movement. According to Snowflake's Snowpark Price Performance: Customer Results report, as of June 2023, customers saw a median of 3.5x faster performance and a 34 percent cost savings with Snowpark

over managed Spark. The performance and price benefits are largely due to these reasons:

- » **Snowflake's distributed compute engine** is logically integrated but physically separated from storage. It was built using a multiclustered, shared data architecture that plans and optimizes the execution of concurrent workloads, with many built-in optimizations such as autoclustering and micro-partitioning.
- » **Snowflake's data platform** enables developers working in Snowpark to bring their code to the data rather than exporting data to other environments. The elimination of external data processing engines for pipelines built in multiple languages — which often requires converting, egressing, scanning, and loading data from its source — helps to reduce costs and improve performance.

Building with Streamlit in Snowflake

Developers need to have as much flexibility in distributing their apps as they do in building them. By using Snowpark as the processing tier for apps, developers also benefit from all the other capabilities in the Snowflake platform. Whether you want to turn ML models into apps, or do everything from build to monetize an app in one environment, Snowflake has options based on your preferences. (See the following section for more about monetization.)

Streamlit in Snowflake enables Python developers to turn data and ML models into interactive web apps — no front-end development required. Streamlit is a pure-Python open-source application framework that enables developers to quickly and easily write, share, and deploy applications using Snowflake as a data source.

Data scientists and developers can

- » **Build** apps using Streamlit's component-rich, open-source Python library.

- » **Modify** code and see changes go live with side-by-side editor and app preview screens in Snowflake.
- » **Share** Streamlit apps via URLs that leverage existing role-based access controls and run on Snowflake's scalable, secure, and performant infrastructure.

Distribute and Monetize with Snowflake Native Apps and Snowflake Marketplace

With Snowflake, developers can build, deploy, distribute, and monetize applications inside the Data Cloud. They can use the Snowflake Native App Framework to bring together all the components of their application, using Snowflake first-party capabilities such as Snowpark and Streamlit in Snowflake. The app can then be distributed and monetized in Snowflake Marketplace and deployed directly inside a customer's Snowflake account. This provides a convenient way for developers to reach a large audience of potential customers and generate revenue from their applications.

Snowflake Marketplace allows developers to distribute and sell their Snowflake-powered applications to other Snowflake customers. From the consumer side, Snowflake customers can discover and purchase Snowflake Native Apps on Snowflake Marketplace, and then install and run them within their own Snowflake account. Bringing the apps close to the data opens up a whole new world of possibilities to enrich, activate, enhance, visualize, and transform data — without data ever leaving the consumer's account.



TECHNICAL
STUFF

Because Snowflake Native Apps run in the customer's account and there is no need to move data, security and procurement hurdles are reduced, which also accelerates sales cycles for providers and time to value for customers. Risk to providers' intellectual property is reduced as the code remains private from end users. Customers can access only the interface visible to them without ever accessing proprietary data sets or logic from providers.



Snowflake offers a range of monetization options, from subscription-based models to usage-based models priced per month or query. Custom Event Billing capabilities enable you to build your own pricing strategy. You can charge customers based on their usage and specify billing events based on your preferences, such as consumed rows, ingested rows, unique ingested users, locations monthly, and more.

Customers see benefits as well: With on-platform billing, customers can pay via credit card, ACH, or even use their Snowflake Capacity commitment. And because these customer-facing apps are Powered by Snowflake, users know they'll receive the benefits of Snowflake's scalability, speed, and reliability.

BUILD, DISTRIBUTE, MONETIZE, AND DEPLOY APPS — ALL IN THE DATA CLOUD

How it works for app providers:

1. **Build** the app's processing tier using Snowpark.
2. **Build** the app's user interface with Streamlit in Snowflake.
3. **Deploy and test** the app locally.
4. **Distribute and monetize** apps on Snowflake Marketplace.

Achieve a competitive edge by building apps on Snowpark

Today's organizations are building modern applications — apps that are data-intensive and AI-enriched — to gain a competitive advantage. But there are major compute challenges involved in processing large volumes of complex and dynamic data from different sources, analyzing it with AI/ML models, and presenting insights so that customers or employees can make better decisions and perform smarter actions. This book reveals how Snowpark leverages the Snowflake Data Cloud to help developers overcome these challenges to build, deploy, and monetize applications quickly, easily, and cost effectively.

Inside...

- What modern applications require
- Using a cloud data platform to develop apps
- Why use Snowflake to build apps
- What you can build with Snowpark
- The benefits of building with Snowpark
- How to get started with Snowpark



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